1(a)	(gravitational) force per unit mass	**B1**
1(b)(i)	$g = GM / r^2$	**C1**
	$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^6)^2$ $= 3.73\ \text{N kg}^{-1}$	**A1**
1(b)(ii)	$a = r\omega^2$ **and** $\omega = 2\pi / T$ **or** $a = v^2 / r$ **and** $v = 2\pi r / T$	**C1**
	$a = 3.39 \times 10^6 \times (2\pi / (24.6 \times 3600))^2$ $= 0.0171\ \text{m s}^{-2}$	**A1**
1(b)(iii)	force per unit mass $= 3.73 - 0.0171$ $= 3.71\ \text{N kg}^{-1}$	**A1**